Techie Academy Capstone Project

Northwind Sales & Business Performance Analysis

Project Scenario

You have been hired as a **Data Analyst for Northwind Traders**, a company that sells a variety of products to customers across different locations.

The company's CEO wants to better understand the company's **sales performance, customers, products, employees, and operations**.

You have been given the company's database in the form of a `.db` file.

Your task is to work as a **Data Analyst**, connect the database to **Power BI**, analyze the data, build an interactive report/dashboard, and present your findings and recommendations to the CEO.

This is not just a visualization exercise.

Your goal is to **turn raw business data into useful information that can support management decision-making.**

Project Objective

Using the provided **Northwind database**, you will:

1. Connect the `.db` database to Power BI.
2. Explore and understand the database.
3. Identify the relationships between the tables.
4. Build an appropriate data model in Power BI.
5. Clean and transform the data where necessary.
6. Create meaningful KPIs and measures.
7. Analyze sales and business performance.
8. Build an interactive Power BI report/dashboard.
9. Identify important business insights.
10. Make data-driven recommendations to the CEO.
11. Present your findings as an executive presentation.
12. Document the entire project on GitHub.

Dataset

You will use the provided:

`Northwind.db`

The database contains information relating to:

- Customers
- Orders
- Order Details
- Products
- Categories
- Suppliers
- Employees
- Shippers
- Territories
- Regions

You are expected to **explore the database yourself** and understand how the tables relate to one another.

Important

Do not simply import every table into Power BI without understanding the database.

Before building your report, investigate:

- What each table represents
- Primary keys
- Foreign keys
- Relationships between tables
- Fact tables
- Dimension tables
- Which tables are necessary for your analysis

Part 1 — Connect the Database to Power BI

Your first task is to figure out how to connect the provided `.db` file to Power BI.

You are expected to research and determine the appropriate method for connecting the database to Power BI.

Document:

- How you connected the database
- What connector/driver you used, if applicable
- Any challenges you encountered
- How you solved those challenges

Learning objective

This is intentionally part of the challenge.

Do not wait for the mentor to provide the connection steps.

A good Data Analyst should be able to investigate unfamiliar data sources and determine how to work with them.

Part 2 — Understand the Data

Before creating visualizations, explore the database.

Answer questions such as:

- How many customers are there?
- How many products are available?
- How many orders were placed?
- How many employees are recorded?
- How many suppliers are there?
- How many product categories exist?
- What countries are represented?
- What period does the order data cover?

You should also identify the important relationships between tables.

Part 3 — Build the Power BI Data Model

Create an appropriate data model in Power BI.

You should be able to identify:

Fact data

For example, the **Order Details** table contains transactional information such as:

- Order
- Product
- Unit Price
- Quantity
- Discount

Dimension data

Potential dimensions include:

- Products
- Customers
- Categories
- Employees
- Suppliers
- Shippers
- Geography/Territories
- Date

Your model should be designed thoughtfully rather than simply creating relationships because Power BI allows them.

You should demonstrate understanding of:

- Primary keys
- Foreign keys
- One-to-many relationships
- Filter direction
- Dimension vs fact tables
- Date tables

Part 4 — Data Cleaning & Transformation

Use **Power Query** where appropriate to prepare the data.

Check for issues such as:

- Incorrect data types
- Missing values
- Duplicate records
- Incorrect date formats
- Unnecessary columns
- Inconsistent text
- Columns that require transformation

Document important transformations.

You do **not** need to transform data simply for the sake of doing so.

The goal is to produce **clean, reliable and analysis-ready data**.

Part 5 — Business Analysis

This is one of the most important parts of the project.

You should use Power BI to investigate the company's performance.

Your analysis should answer meaningful business questions.

A. Overall Business Performance

Create KPIs such as:

- Total Sales/Revenue
- Total Orders
- Total Quantity Sold
- Average Order Value
- Number of Customers
- Number of Products
- Number of Categories

You may create additional KPIs where they provide business value.

B. Sales Performance

Investigate:

- How are sales changing over time?
- Which months/years generated the most sales?
- Which products generate the most revenue?
- Which categories perform best?
- Which countries generate the most sales?
- Which customers contribute the most revenue?

C. Product Performance

Analyze:

- Top-selling products
- Products generating the highest revenue
- Best-performing categories
- Products with low sales
- Products with high inventory
- Discontinued products
- Product pricing

Consider whether there are products that management should:

- Promote
- Reprice
- Monitor
- Restock
- Discontinue

Your recommendations should be based on evidence from the data.

D. Customer Analysis

Investigate:

- Number of customers
- Sales by country
- Top customers
- Customer contribution to revenue
- Customer purchasing patterns

Identify the company's most valuable customer segments or locations.

E. Employee/Sales Performance

Analyze the sales performance associated with employees.

Consider:

- Sales by employee
- Number of orders handled
- Revenue contribution
- Average order value
- Performance over time

Be careful when interpreting employee performance.

For example, **higher sales do not automatically mean an employee is more productive**. Consider the number of orders, customers served, time period and other relevant factors.

F. Shipping & Operations

Investigate the company's shipping activity.

For example:

- Orders by shipper
- Freight costs
- Shipping patterns
- Orders shipped on time vs late, if the data supports this analysis
- Shipping performance by country

Identify any operational patterns that management should be aware of.

Part 6 — Build the Power BI Report

Your Power BI file should contain a **professional multi-page report**.

A suggested structure is:

Page 1 — Executive Overview

This should be the CEO's starting point.

Include:

- Key KPIs
- Overall sales performance
- Sales trend
- Top products/categories
- Geographic performance
- Important highlights

The CEO should be able to understand the company's overall performance within a few seconds.

Page 2 — Sales Analysis

Include visualizations such as:

- Sales trend
- Sales by category
- Sales by country
- Sales by year/month
- Top products
- Top customers

Page 3 — Product Performance

Analyze:

- Product sales
- Product quantity
- Category performance
- Top/bottom products
- Inventory-related insights where appropriate

Page 4 — Customer & Geographic Analysis

Include:

- Sales by country
- Customer distribution
- Top customers
- Customer contribution to revenue

Page 5 — Employee & Operations

Include relevant analysis of:

- Employee performance
- Orders
- Shippers
- Freight
- Shipping performance

You **do not have to follow these pages exactly**.

If your analysis suggests a better structure, create your own.

Dashboard Design Guidelines

Your dashboard should be:

Clear

Someone should understand what each visualization communicates.

Professional

Use consistent:

- Fonts
- Formatting
- Number formats
- Titles
- Spacing
- Layout

Interactive

Where appropriate, include:

- Slicers
- Filters
- Drill-through
- Tooltips
- Cross-filtering
- Navigation buttons

Avoid visualization overload

Do not put 15–20 charts on one page just because you can.

Every visual should answer a business question.

Part 7 — Insights

This is where you move from **Data Analyst → Business Analyst**.

Do not simply say:

> "The USA had the highest sales."

Explain what that means for the business.

For example:

> "The USA generates the largest share of revenue, suggesting that it is a key market for Northwind. Management should consider prioritizing customer retention and marketing efforts in this market."

Your insights should answer:

What happened? → Why does it matter? → What should the business do?

Aim for approximately **5–10 strong insights** rather than listing dozens of observations.

Part 8 — Recommendations to the CEO

Based on your analysis, provide practical recommendations.

Your recommendations could relate to:

- Products
- Customers
- Marketing
- Pricing
- Inventory
- Sales strategy
- Geographic markets
- Employee performance
- Shipping/operations

Important:

Your recommendations must be **supported by your analysis**.

Do not make recommendations simply because they sound good.

Part 9 — CEO Presentation

Create a presentation for the CEO.

Your presentation should **not be a copy of your Power BI report**.

Imagine that you have **10–15 minutes with the CEO** and need to explain what the data is telling them.

Suggested presentation structure

Slide 1 — Title

Northwind Business Performance Analysis

Your name
Data Analyst

Slide 2 — Executive Summary

Briefly explain:

- What you analyzed
- The major business findings
- The overall performance

Slide 3 — Business Performance

Present the most important KPIs and overall performance.

Slide 4 — Sales Performance

Explain the major sales trends.

Slide 5 — Product Performance

Highlight important product/category findings.

Slide 6 — Customer & Market Performance

Explain customer and geographic insights.

Slide 7 — Operational/Employee Insights

Present relevant operational findings.

Slide 8 — Key Business Insights

Present your strongest findings.

Slide 9 — Recommendations

Give the CEO clear actions they should consider.

Try to make your recommendations specific.

Instead of:

> "The company should improve sales."

Say:

> "Management should prioritize the highest-performing markets while developing targeted strategies for underperforming regions."

Slide 10 — Conclusion

Summarize:

- What the data tells us
- What management should focus on
- The expected business value of acting on the recommendations

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Part 10 — GitHub Submission

Create a **public GitHub repository** for your capstone project.

Use a professional repository name, for example:

`northwind-business-analysis`

or

`northwind-sales-analysis`

Your repository should contain:

```
northwind-business-analysis/
│
├── README.md
│
├── data/
│   └── Northwind.db
│
├── powerbi/
```

```
│   └── Northwind_Analysis.pbix
│
├── pdf/
│   └── Northwind_Report.pdf
│
├── presentation/
│   └── Northwind_CEO_Presentation.pptx
│
├── screenshots/
│   ├── executive-dashboard.png
│   ├── sales-analysis.png
│   ├── product-analysis.png
│   └── customer-analysis.png
│
└── documentation/
    └── data-model.png
```

Important

You **do not necessarily need both the `.pbix` and PDF**.

However:

Preferred: Submit the `.pbix` file **and** a PDF export of the report if possible.

If you publish your report to **Power BI Service**, you may also include the Power BI Service link where appropriate.

Part 11 — Your README.md

Your GitHub README should professionally document the project.

It should contain:

- Project title
- Project overview
- Business problem
- Objectives
- Dataset description
- Database structure
- Tools used

- Data preparation
- Data model
- Analysis performed
- Dashboard/report preview
- Key insights
- Recommendations
- Project structure
- How to access/use the project
- Author information

Include screenshots of your **best Power BI pages** in the README.

For example:

Dashboard Preview

![Executive Dashboard](screenshots/executive-dashboard.png)

Your README should allow someone visiting your GitHub repository to understand your project **without opening the Power BI file**.

Final Submission Checklist

Before submitting, make sure your GitHub repository contains:

Dataset

- `Northwind.db`

Power BI

- `.pbix` Power BI file
- PDF export of report, if available
- Power BI Service link, if applicable

Presentation

- CEO presentation `.pptx`
- Presentation explains findings
- Presentation includes recommendations

Documentation

- Professional `README.md`
- Dashboard screenshots
- Data model/ERD screenshot
- Key insights
- Recommendations

Quality Check

- No broken links
- No missing images
- Dashboard is readable
- Visualizations have meaningful titles
- Numbers are correctly formatted
- Recommendations are supported by data
- Files are properly organized
- GitHub repository is accessible

Final Submission

When you have completed the project, submit **your GitHub repository link**.

Submission format:

Name: Your Full Name
GitHub Repository:
`https://github.com/yourusername/northwind-business-analysis`
Power BI Service: [Optional link]

What This Capstone Is Testing

This project is designed to assess your ability to work through a **real-world analytics workflow**.

You will be assessed on:

Area	What is being assessed
Database	Understanding and working with a `.db` database
Data Connection	Ability to connect an unfamiliar data source to Power BI
Data Modeling	Understanding relationships and dimensional modeling
Data Preparation	Cleaning and transforming data appropriately
DAX/KPIs	Creating meaningful business measures
Analysis	Ability to investigate business questions
Visualization	Choosing appropriate and effective visuals
Dashboard	Design, usability and interactivity
Insights	Ability to turn numbers into meaningful findings
Recommendations	Ability to translate findings into business actions
Presentation	Ability to communicate findings to an executive
GitHub	Professional project documentation and organization

The mindset for this project

Don't approach this as:

> "I need to create some charts in Power BI."

Approach it as:

> **"The CEO has given me access to the company's data. What does the data tell me, what problems can I identify, and what should the business do next?"**

That mindset is what turns a **Power BI assignment into a Data Analytics capstone project.**